EST II — MATH LEVEL 2 • UNIT 7.01

C Descriptive Statistics, Normal Distribution & Regression

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Descriptive Statistics, Normal Distribution & Regression Practice Exercises

1 MCQ

If $i = \sqrt{-1}$, which of the following is equivalent to the complex expression $\frac{7+4i}{2-i}$?

A $2 + 3i$

B $2 - 3i$

C $\frac{10+15i}{3}$

D $\frac{18+15i}{5}$

2 MCQ

The graph of the function $y = f(x)$ is shown below. What is the value of $f(1)$?

A 0

B 2

C -1

D 3

3 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?

A (1, 0) and (3, 0)

B (0, 1) and (0, 3)

C (-1, 0) and (-3, 0)

D (1, 0) and (-3, 0)

4 MCQ

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

A [3, 4]

B [5, 8]

C [-3, -4]

D [4, 3]

5 MCQ

A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?

A $\frac{3}{5}$

B $\frac{2}{5}$

C $\frac{1}{2}$

D $\frac{3}{4}$

6 MCQ

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

A 14

B 17

C 8

D 11

7 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = 2$. Find a_5.

A 96

B 192

C 98

D -96

8 MCQ

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.

A 0

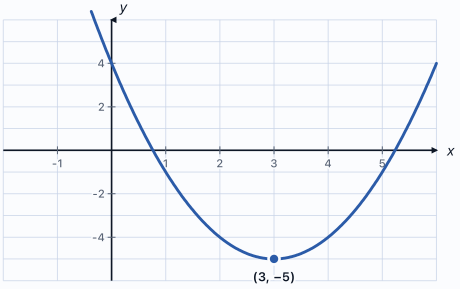
B -2

C 2

D 1

9 MCQ

The graph of the function $f(x) = x^2 - 6x + k$ is shown below. The graph has a vertex at $(3, -5)$. Determine the value of the parameter k.



A 9

B -4

C 4

D 1

10 MCQ

A translation maps point $A(1, 2)$ to $A'(-4, 2)$ and point $B(3, 5)$ to $B'(-2, 5)$. This transformation is represented by $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}$. Find the values of a and b.

A a = 0, b = 5

B a = -5, b = 0

C a = 5, b = 0

D a = 0, b = -5

11 MCQ

A survey was conducted on a group of 100 students regarding their preferences for two subjects, Mathematics (M) and Science (S). The results are summarized in the two-way table below, where some values are unknown. If a student is chosen at random from this group, the probability that the student prefers Mathematics but not Science, $P(M \cap \neg S)$, is equal to the probability that the student prefers Science but not Mathematics, $P(\neg M \cap S)$. What is the value of x?

	Science (S)	Not Science (¬S)	Total
Mathematics (M)	25	x	45
Not Math (¬M)	y	35	55
Total	45	z	100

A 30

B 15

C 20

D 25

12 MCQ

Consider a geometric sequence with positive common ratio $r > 0$ where the first term is $a_1 = \frac{1}{2}$. If the sum of the first three terms is $S_3 = \frac{7}{2}$, find the value of r.

A -2

B 2

C 1/2

D 3

13 MCQ

A geometric sequence has $a_1 = 2$ and common ratio $r = 3$. Find a_5.

A 162

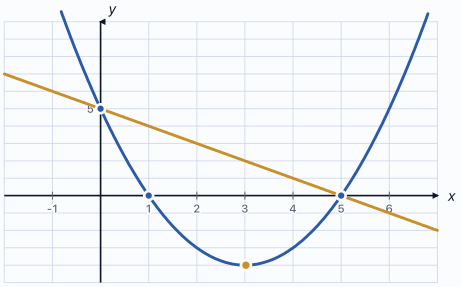
B 486

C 165

D -162

14 MCQ

Consider the functions $f(x) = x^2 - 6x + 5$ and $g(x) = -x + 5$. The graphs of these two functions intersect at two points. Let these points be A and B. A third function, $h(x)$, is defined as the difference between the x-coordinates of these intersection points, i.e., $h(x) = |x_A - x_B|$. What is the value of $h(x)$?



A $\sqrt{25}$

B $\sqrt{16}$

C $\sqrt{36}$

D $\sqrt{49}$

15 MCQ

In an applied problem, a geometric sequence has $a_1 = 6$ and common ratio $r = 3$. Find a_5.

A 1458

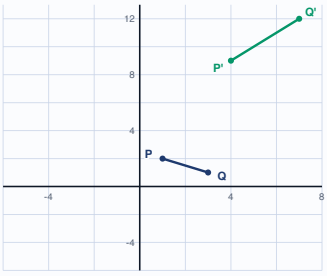
B 489

C -486

D 486

16 MCQ

A linear transformation in the 2D plane is represented by the matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. This transformation maps the point $P(1, 2)$ to $P'(-3, 7)$ and the point $Q(3, 1)$ to $Q'(5, 1)$. Determine the values of a, b, c, and d, and then calculate the sum $a + b + c + d$.



A 11

B 7

C 9

D 5

17 MCQ

A geometric sequence has $a_1 = 2$ and common ratio $r = -2$. Find a_4.

A 32

B -16

C -18

D 16

18 MCQ

Suppose a geometric sequence has $a_1 = 5$ and common ratio $r = 3$. Find a_3.

A 135

B 48

C -45

D 45

19 MCQ

Consider the complex number $z = 3 + 4i$. What is the magnitude of z?

A 7

B 5

C 12

D 25

20 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What are the coordinates of the vertex of the parabola represented by this function?

A (2, 1)

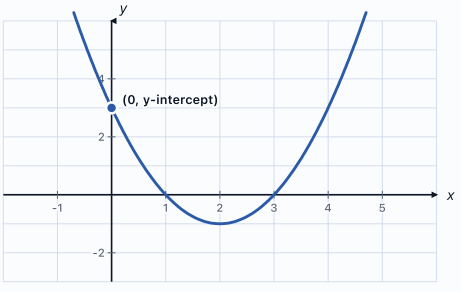
B (1, 0)

C (3, 0)

D (2, -1)

21 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What is the y-intercept of the graph of this function?



A -3

B 4

C -4

D 3

22 MCQ

A geometric sequence has $a_1 = 1$ and common ratio $r = -3$. Find a_5.

A 78

B 81

C -81

D -243

23 MCQ

In an applied problem, a geometric sequence has $a_1 = 1$ and common ratio $r = -3$. Find a_6.

A -243

B -246

C 243

D 729

24 MCQ

An arithmetic sequence has its first term $a_1 = 7$ and a common difference $d = 5$. What is the value of the 3rd term, a_3?

A 12

B 17

C 21

D 35

25 MCQ

Let the complex number w be defined as $w = (a + bi) + (3 - 2i)$, where a and b are real numbers. Given that $|w|^2 = 25$ and w lies in the first quadrant, find the value of $a + b$.

A 5

B 7

C 6

D 8